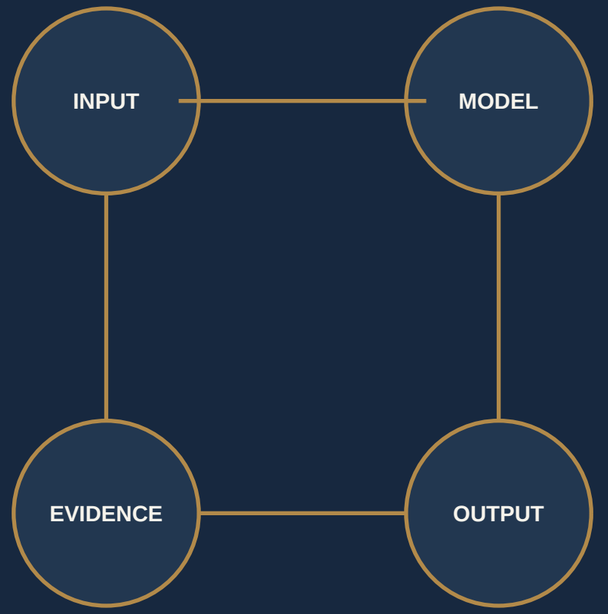
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Vision

AI.410 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define advanced vision specifically for robotic perception.
- Apply key methods including including object detection.
- Build or analyze a practical system involving pose estimation.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Advanced vision specifically for robotic perception

02

METHODS

Including object detection

03

SYSTEMS

Pose estimation

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Advanced vision specifically f

Advanced vision specifically for robotic perception

02 · Including object detection

Including object detection

03 · Pose estimation

Pose estimation

04 · Depth perception

Depth perception

05 · 3D reconstruction

3D reconstruction

06 · Tracking and visual servoing

Tracking and visual servoing

07 · Mandatory practical: Build a v

Mandatory practical: Build a vision-guided robotic system capable of detecting and interacting with objects

Core ideas, made explicit

AI.410 • Distinguish definitions, assumptions and evidence

- Advanced vision specifically for robotic perception
- Including object detection
- Pose estimation
- Depth perception
- 3D reconstruction

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.410 • Distinguish definitions, assumptions and evidence

- Tracking and visual servoing
- Mandatory practical: Build a vision-guided robotic system capable of detecting and interacting with objects

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Advanced vision specifically for robotic perception

INPUTS

Including object detection

METHOD

Pose estimation

OUTPUT

Depth perception

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving advanced vision specifically for robotic perception.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using advanced vision specifically for robotic perception.
- Implement or simulate a baseline using including object detection.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.